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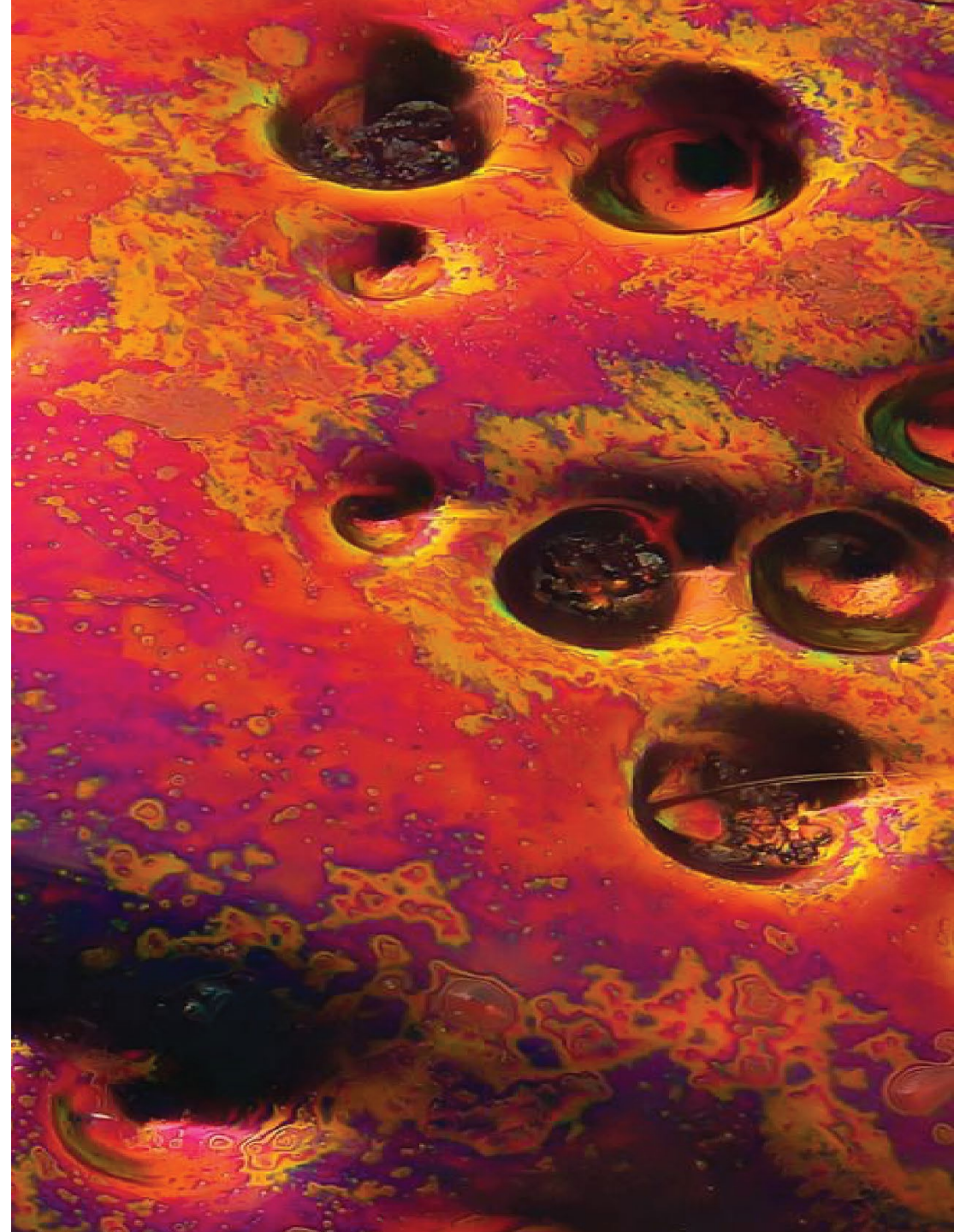
School of Clinical Medicine
Department of Ophthalmology
香港大學眼科學系

The Red Eye

Dr Allie Lee
Clinical Assistant Professor

aleeni@hku.hk

WCS 24 Sep 2022





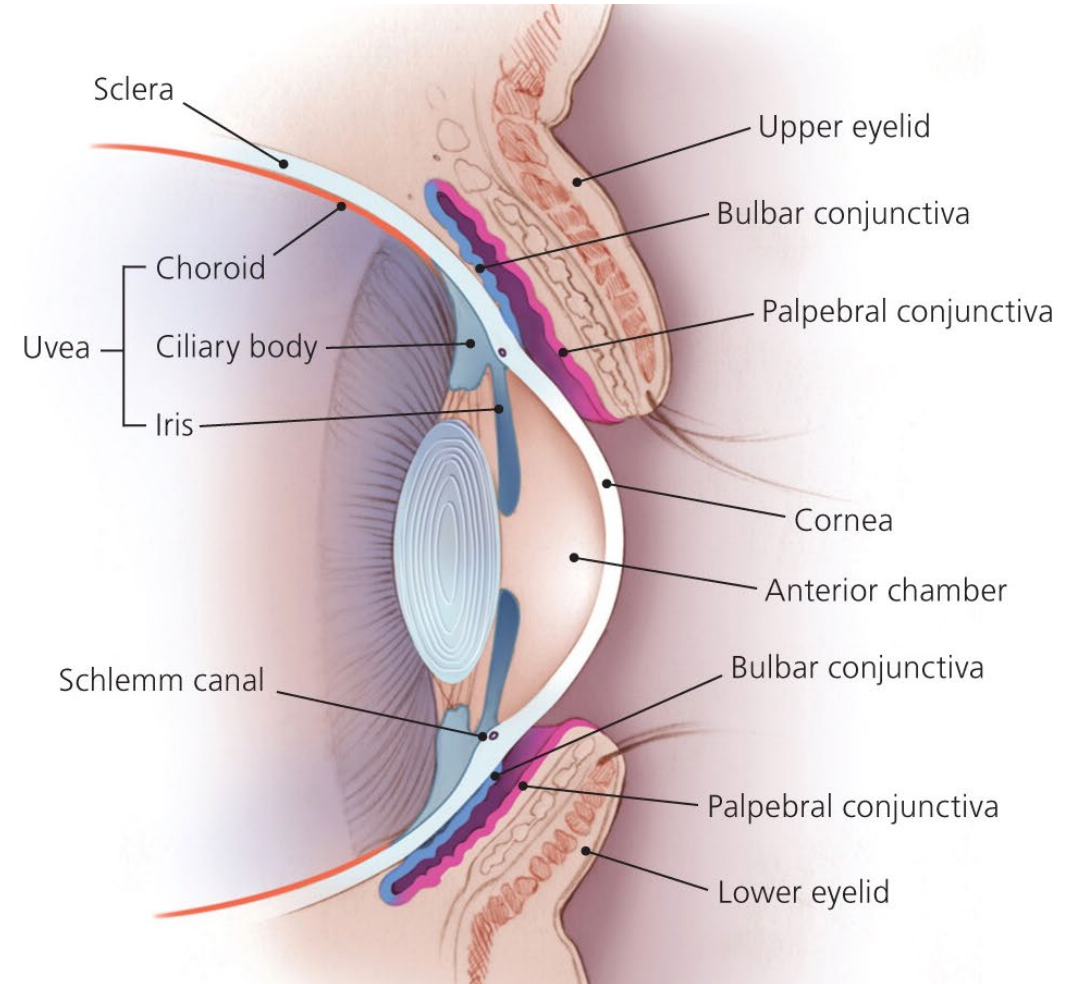
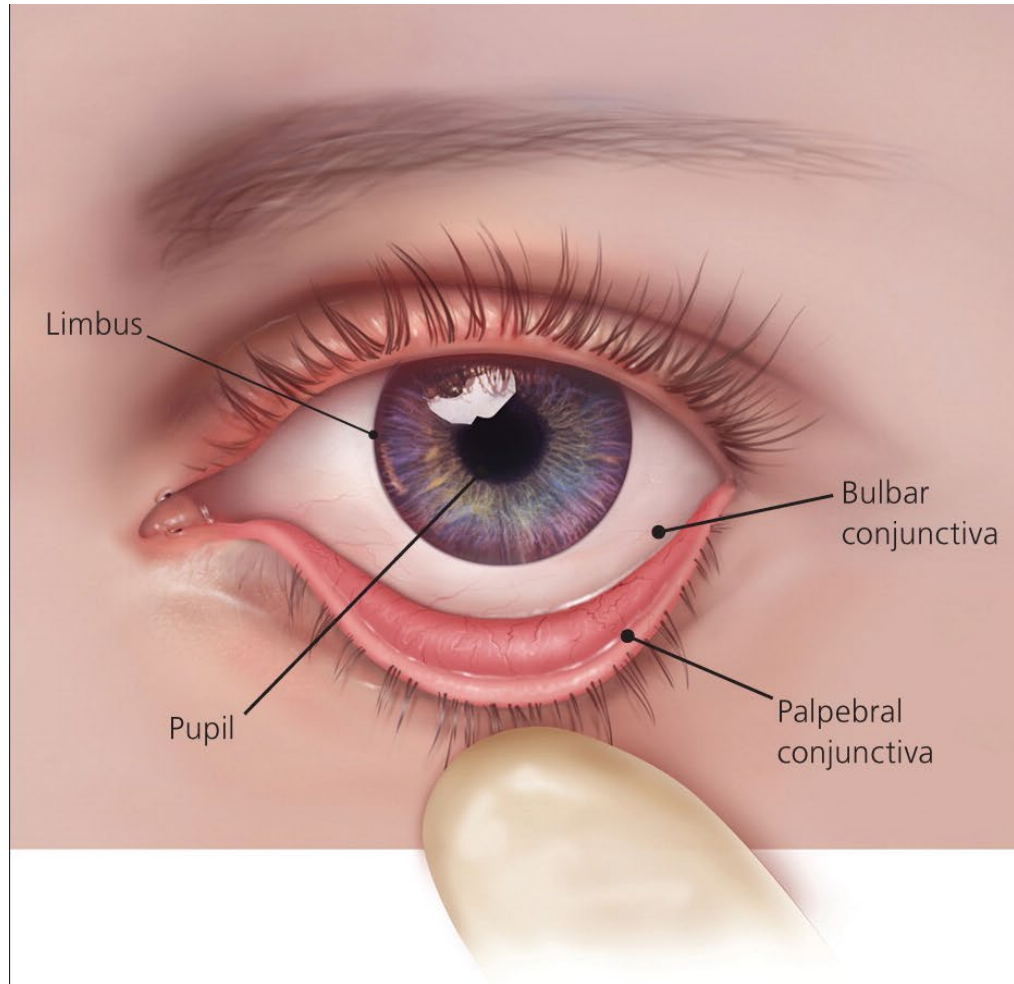
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Learning Objectives

- Familiarize with a range of *common* ocular and external eye diseases
- Describe a framework for diagnosis and treatment
- Learn the skills to promptly recognize and manage the red eye emergencies

The Ocular Surface





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History

- *Redness, pain, blurred vision*
- Onset, recurring
- Risk factors: trauma, recent surgery, contact lens use, ophthalmic medications
- Systemic disease: rheumatological diseases, thyroid

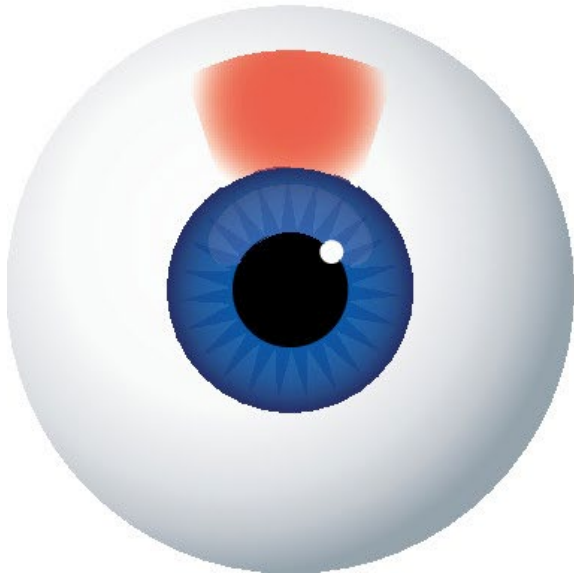


Eye examination for non-ophthalmologists

- Visual acuity
- Pupil: size, reflexes, RAPD
- External eye: lid abnormality, ptosis
- Inspection
 - Characteristics of redness: colour (bright/dull, red/pink), distribution (ciliary, sectoral, diffuse),
 - Associated signs: cornea opacity, mass, discharge

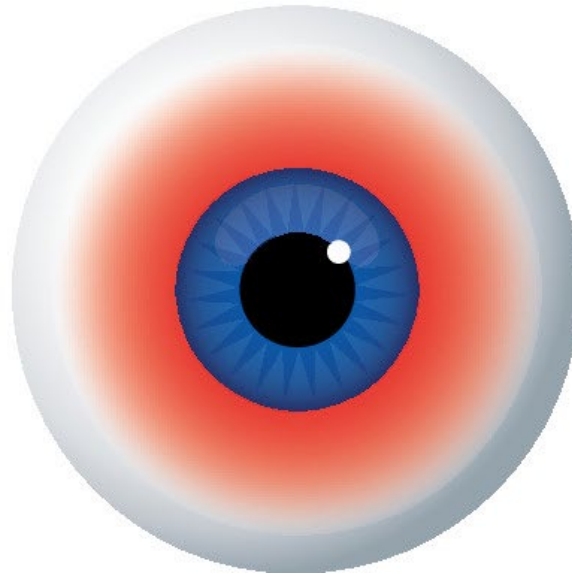
Distribution of redness

Sectorial injection



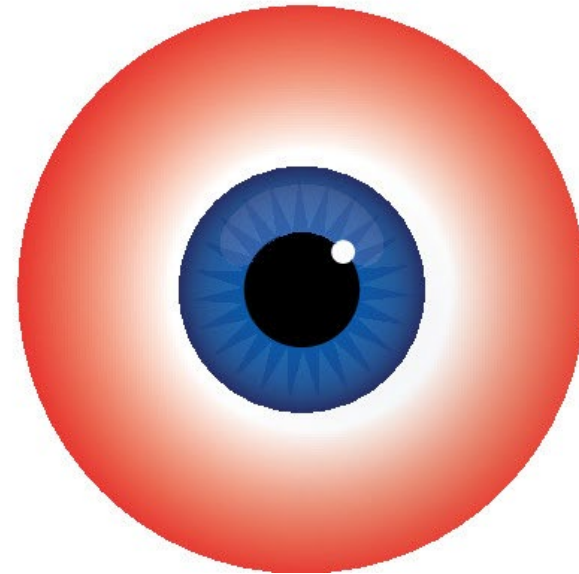
SCH, scleritis,
episcleritis, trauma

Circumferential injection



Uveitis, glaucoma

Diffuse injection



Conjunctivitis, drug toxicity,
severe keratitis, endophthalmitis



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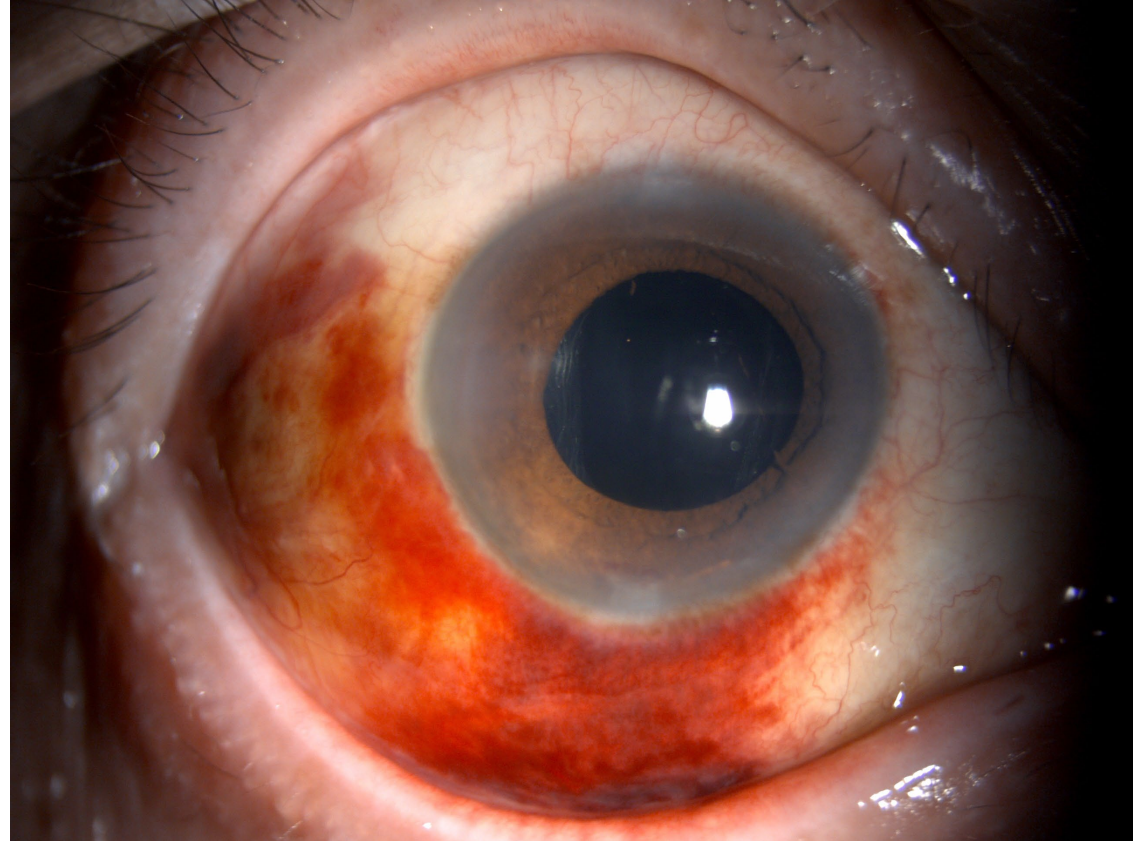
Case 1 60/M

OD redness x 2/7

- no pain, BOV or trauma

PMHx

- DM, HT, lipids
- IHD on aspirin





Subconjunctival haemorrhage

- Characteristic bright homogenous red, asymptomatic
- From ruptured subconjunctival blood vessel → accumulation of blood between conjunctiva and episclera
- Causes
 - Mostly spontaneous and sporadic
 - Trauma: minor (e.g. contact lens), post surgery, eye/head injury
 - Valsalva (straining, coughing, vomiting)
 - Systemic: HT, DM, antiplatelet/anticoagulants, coagulopathy
- Self-limiting; consider work-up if recurrence



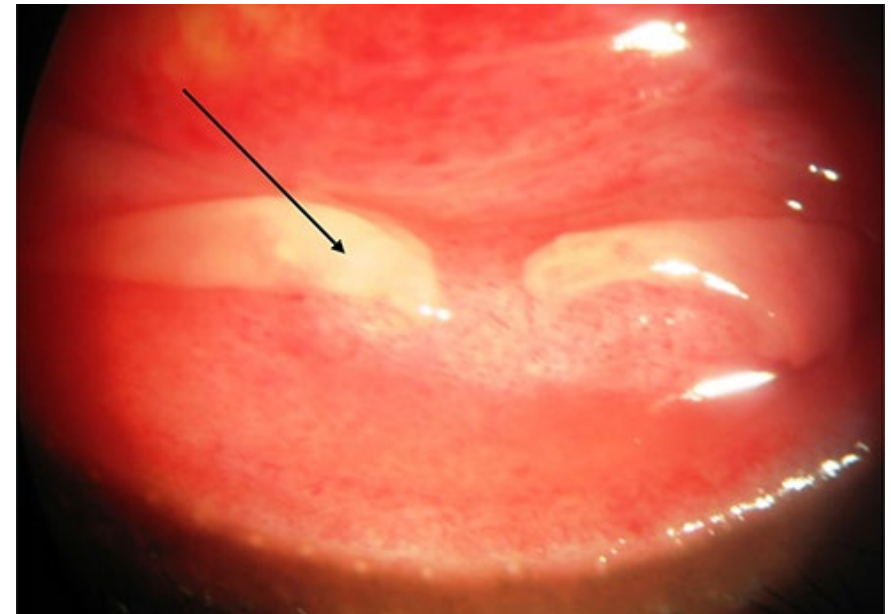
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Case 2 40/F

OU redness x 1/52

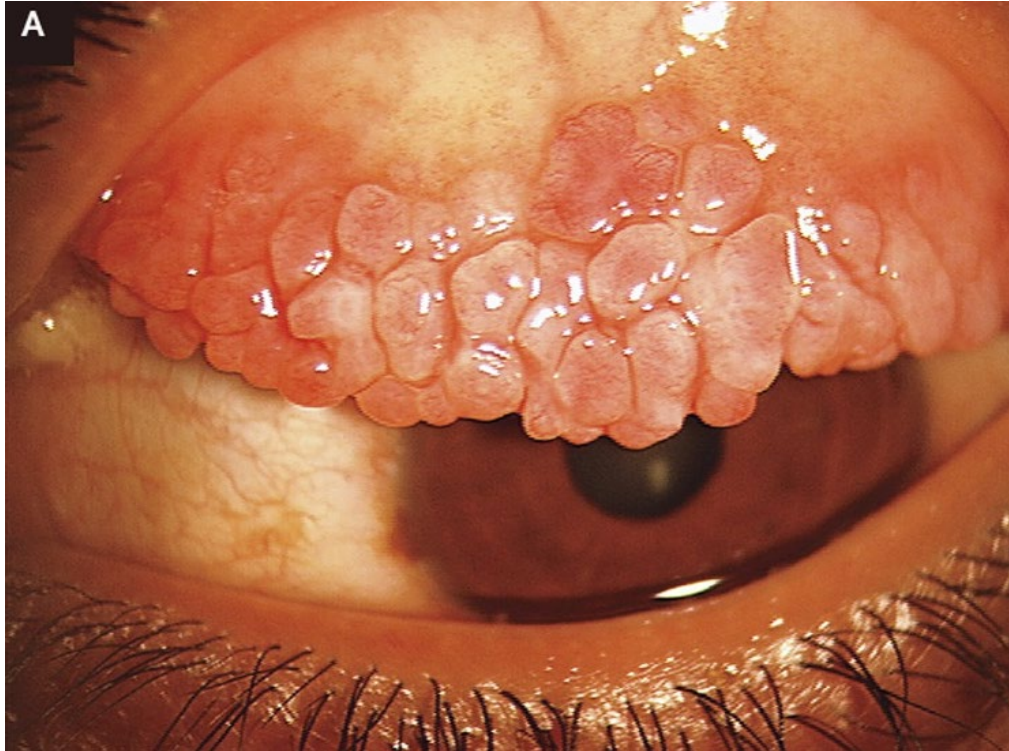
- Itch and discharge, crusting in the morning
- No pain or BOV
- Recent URTI+
- Good past health





Types of Conjunctivitis

	Allergic	Viral	Bacterial
Itchiness	+++	+	+/-
Discharge	Watery White mucus	Serous Morning crusting	Purulent
Features	Papillae	Follicles	-
Systemic	Atopy Contact allergens	URTI	-
Common pathogen(s)	-	Adenovirus	Staphylococcus Streptococcus Haemophilus
Treatment	Anti-histamines Avoid allergens	Self-limiting Contact precaution	Most self-limiting if mild Topical antibiotics



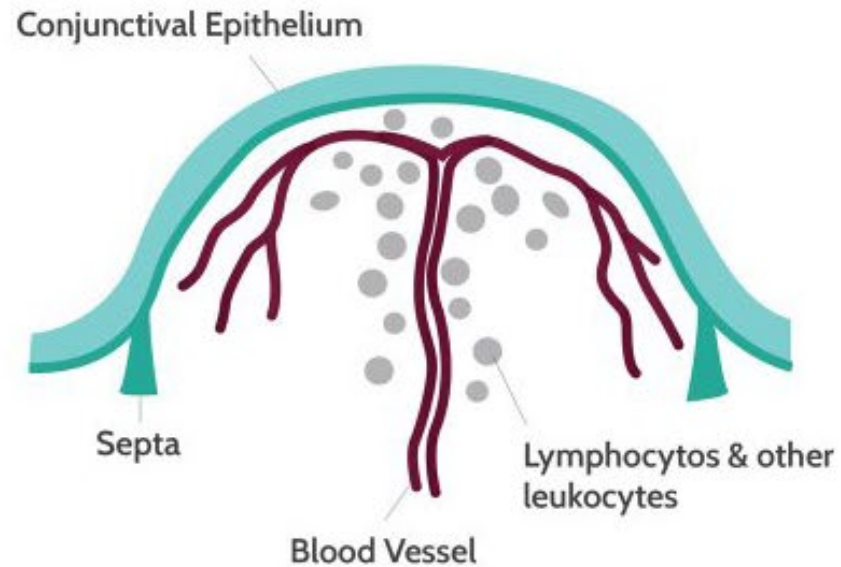
Papillae

- Larger projections with central vascular core
- Allergic immune response due to atopy (VKC, AKC) or foreign body (e.g. contact lens, ocular prosthesis)



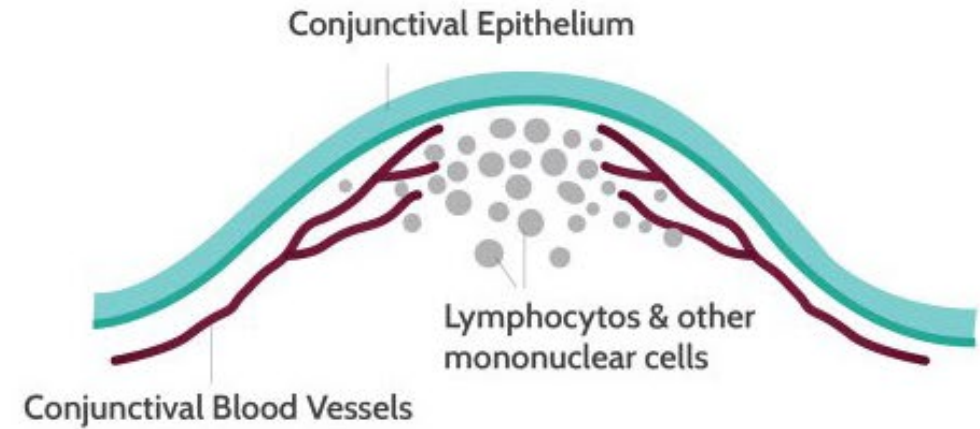
Follicles

- Small, dome-shaped nodules surrounded by conjunctival blood vessels (red base)
- Inflammation by pathogens, e.g. virus, atypical bacteria or toxins (including eyedrops)



Papillary Conjunctivitis

Vascular core surrounded by inflammatory cells



Follicular Conjunctivitis

Lymphoid follicle (germinal center) surrounded by conjunctival vessels



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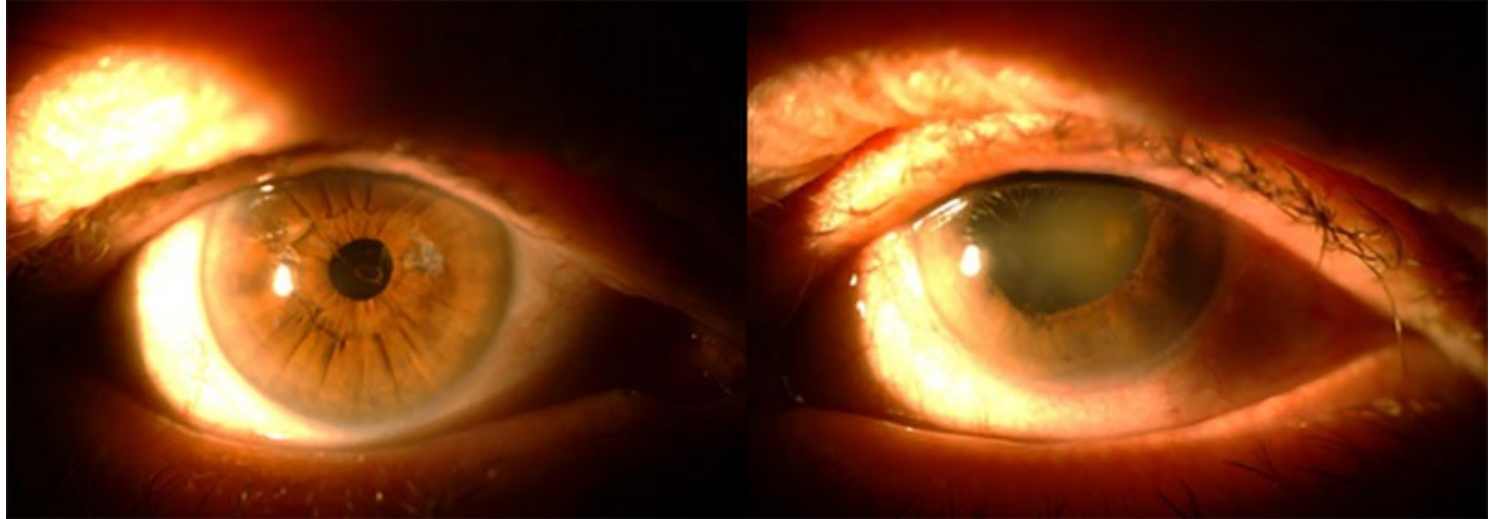
Case 3 60/F

OS redness since this evening

- Pain++, vomiting+
- BOV+

Past health

- Hyperope +4D
- Common cold meds+





Acute primary angle closure: symptoms

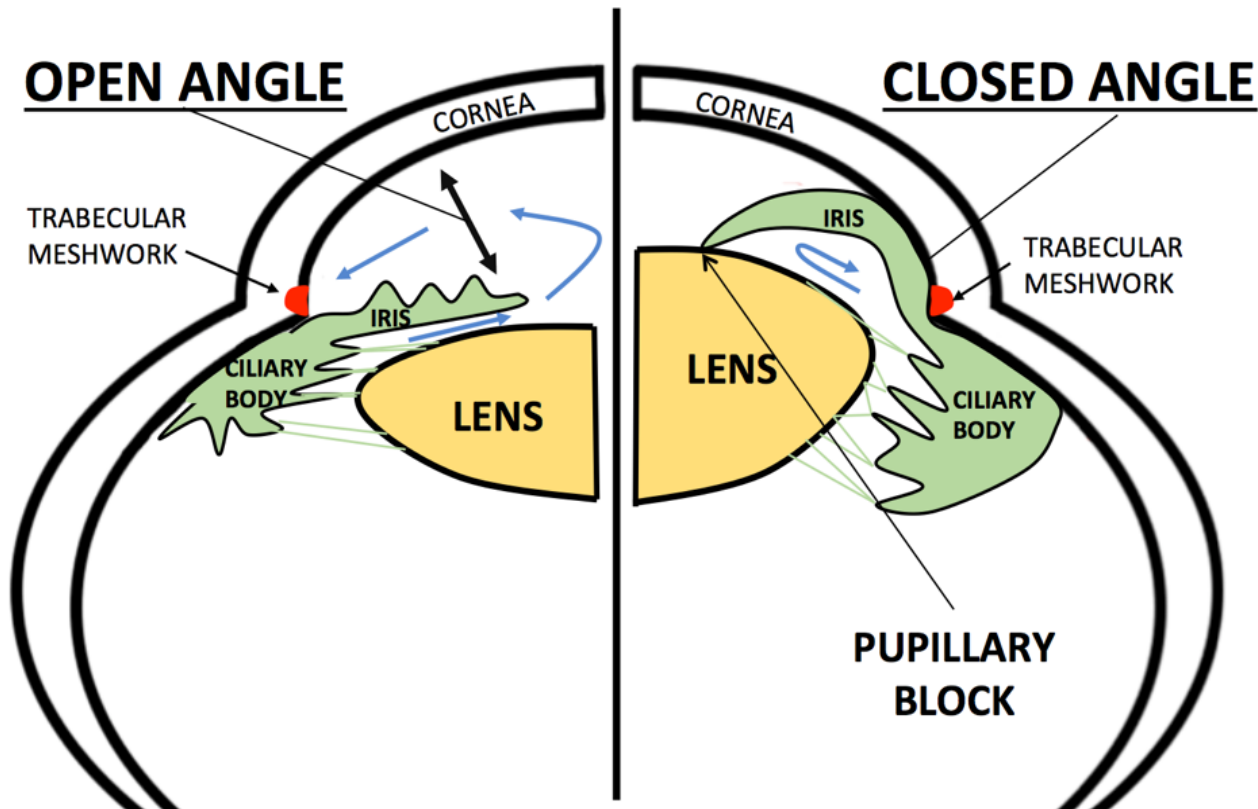
- Severe ocular pain
- Frontal headache
- Blurring of vision
- Halos around lights
- Nausea +/- vomiting

Acute primary angle closure: signs

- Fixed mid-dilated pupil
- Corneal haze
- Shallow anterior chamber
- Conjunctival injection
- *Very high intraocular pressure*



Mechanism: pupil block



Predisposing factors:

Short axial length

Thick lens

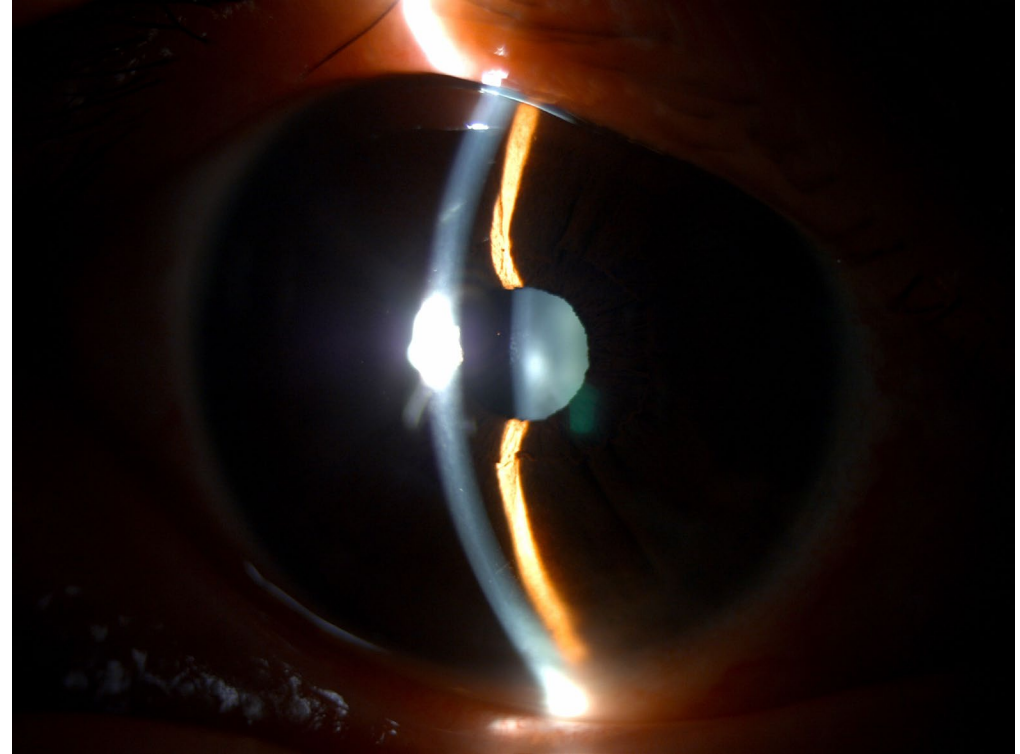
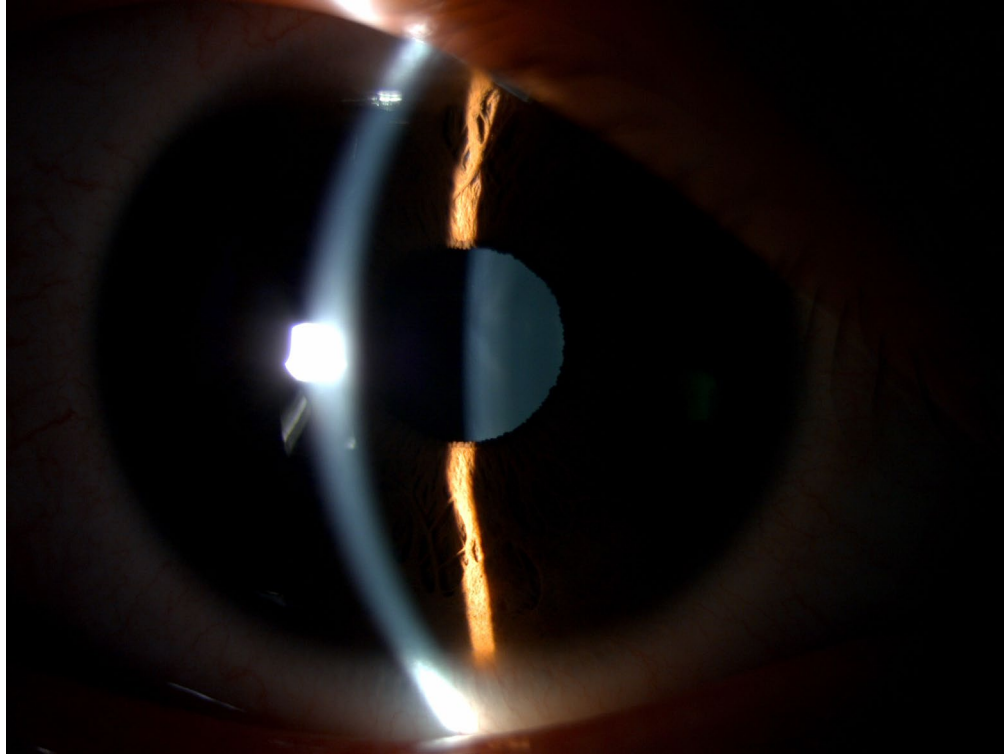
Pupil mid dilatation

(e.g. dim light, dilating eyedrops,
systemic antihistamine or sinus
decongestants)



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Immediate treatment

To miose and medically abort pupil block

- Pilocarpine 4% (cholinergic-agonist)

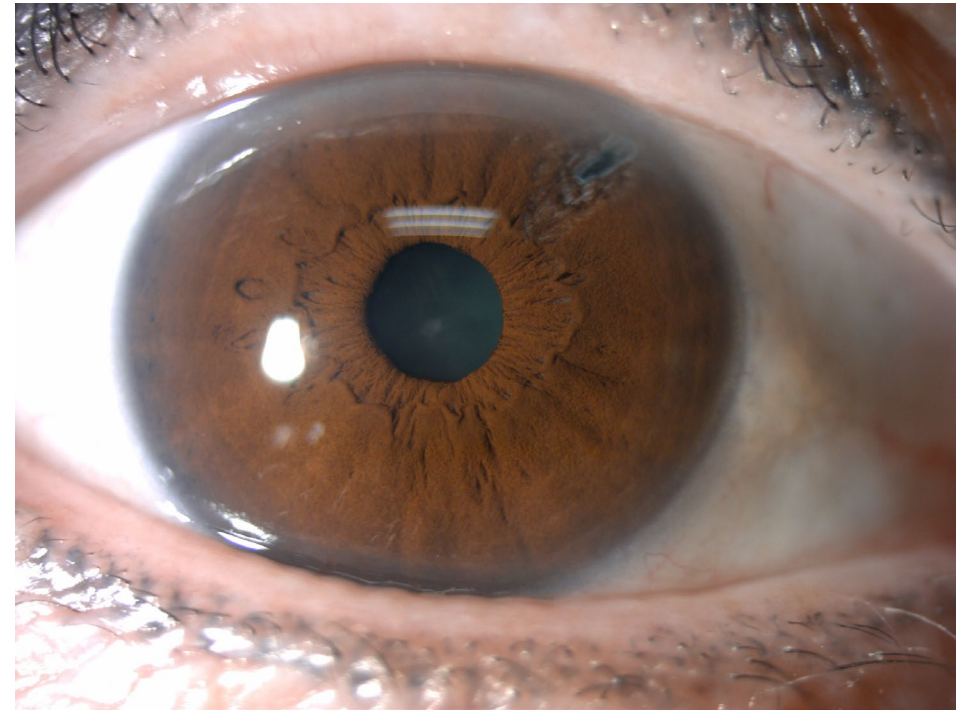
To decrease intraocular pressure

- Timolol 0.5% (beta-blocker)
- Apraclonidine 0.5% (alpha-2-adrenergic agonist)
- IV acetazolamide 500mg (carbonic anhydrase inhibitor)
- IV mannitol 20% 200ml (hyperosmotic agent)



Definitive treatment

- Peripheral iridotomy: provide alternative pathway for aqueous flow
- Lens extraction: to open up angle
- Fellow eye: prophylaxis

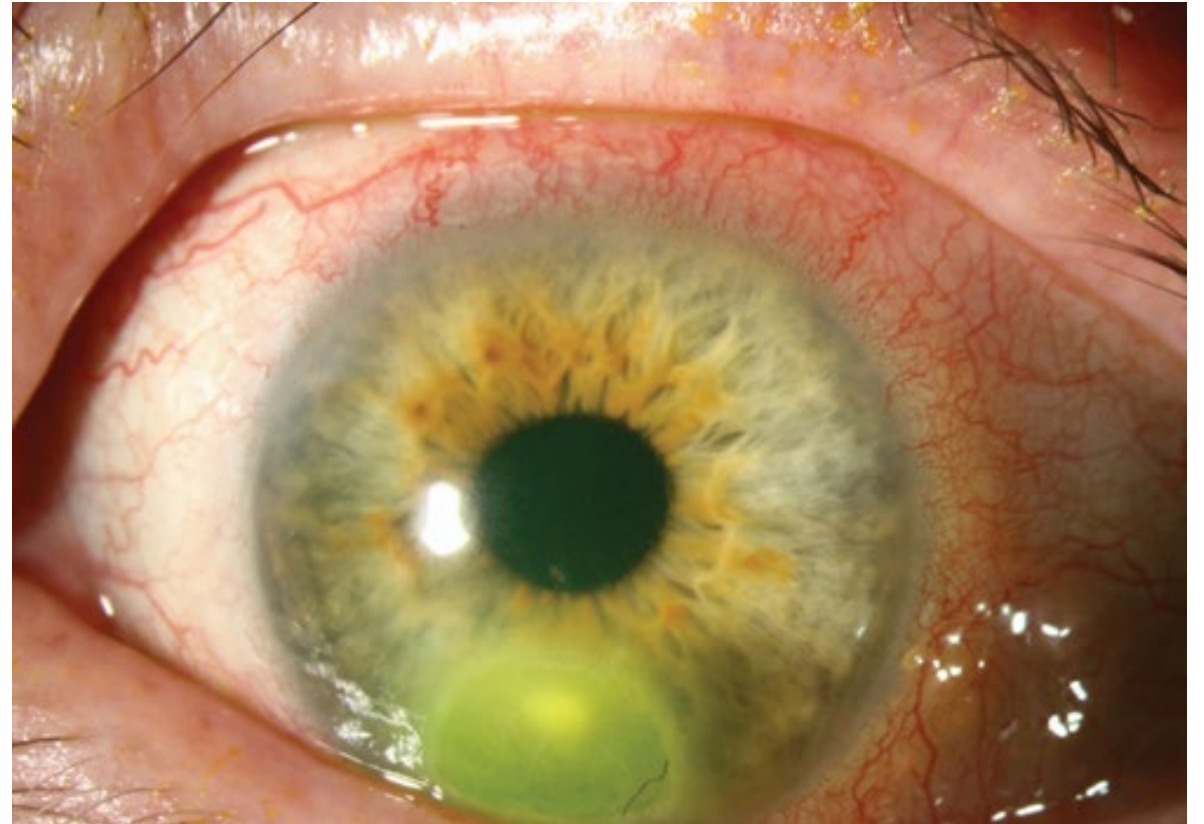




Case 4 20/F

OS redness x 1/7

- Pain+++ BOV+
- Long term contact lens user:
one-month disposable,
overnight wear, swim with CL





Management of microbial keratitis

1. To identify the causative micro-organism

- Corneal scraping of infiltrate for culture and sensitivity testing
- Contact lens, contact lens case, solution

2. To start empirical antibiotics

- Intensive: up to q1h round the clock
- Requires both Gram+ and Gram- coverage
- Monotherapy (e.g. moxifloxacin, levofloxacin 1.5%)
- Combination (e.g. fortified vancomycin + gentamycin/ceftazidime)



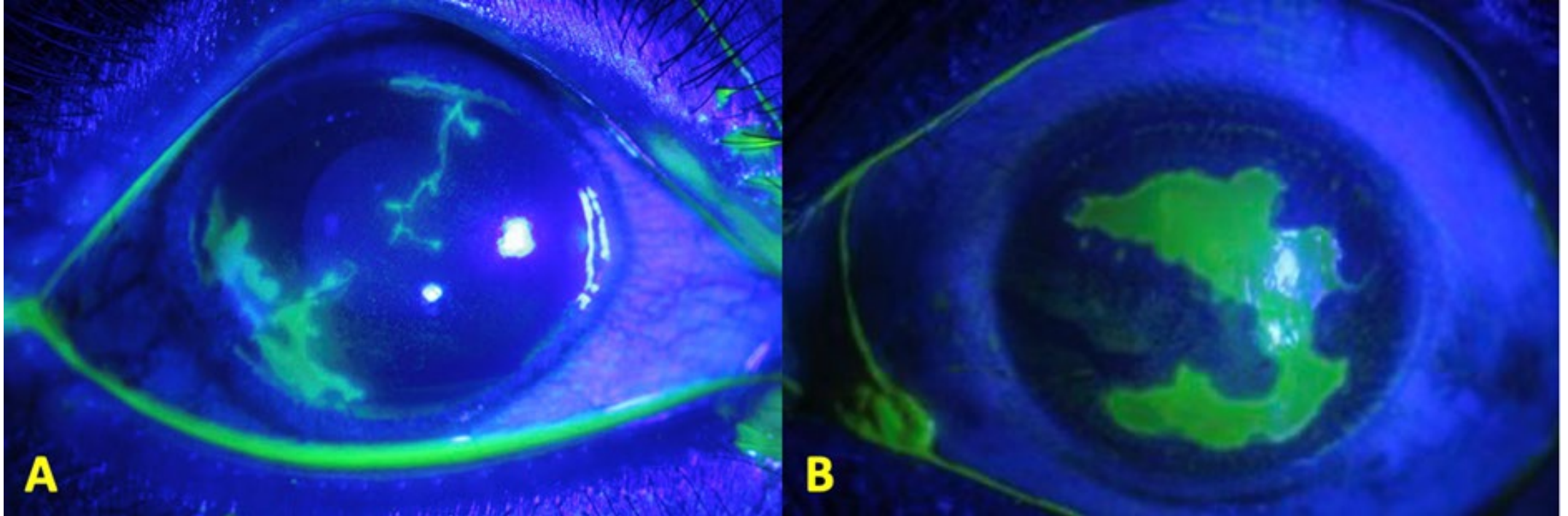
Organisms by risk factors

- CL-related: pseudomonas, acanthamoeba
- Vegetative matters: fungus
- Common: normal skin flora (staphylococcus, coagulase-negative streptococcus)



Herpetic keratitis

- Mostly herpes simplex virus (HSV)
- A clinical diagnosis – laboratory testing rarely needed
- Antiviral agents
 - Topical (acyclovir ointment, ganciclovir gel)
 - Oral (acyclovir, valacyclovir)



HSV ulcers: dendritic vs geographic

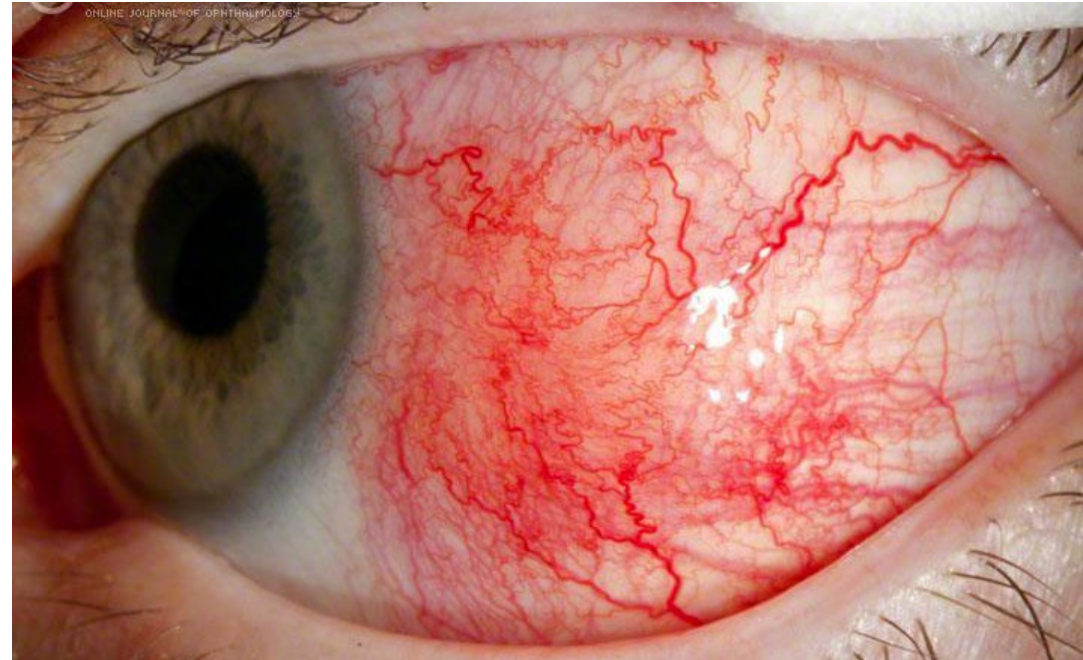
Case 5 40/M

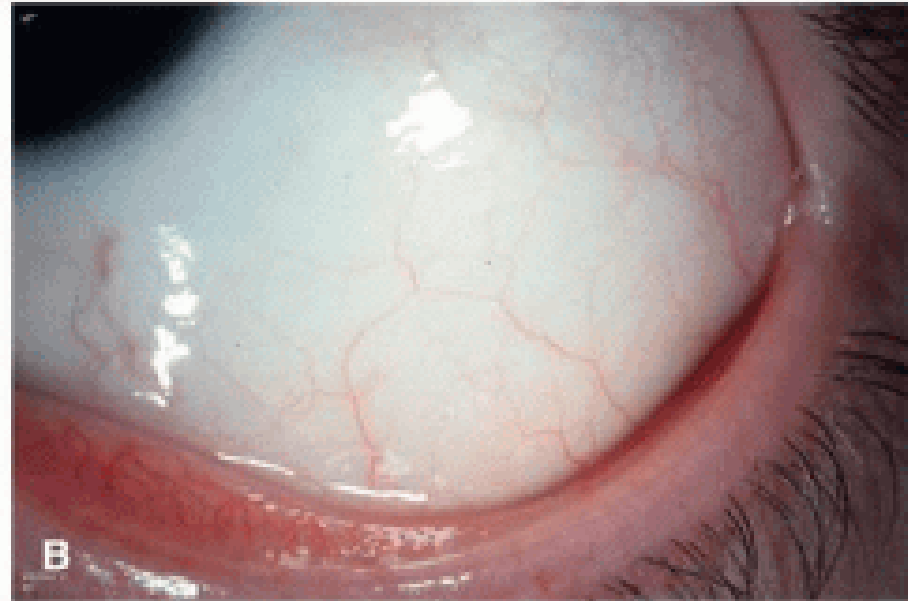
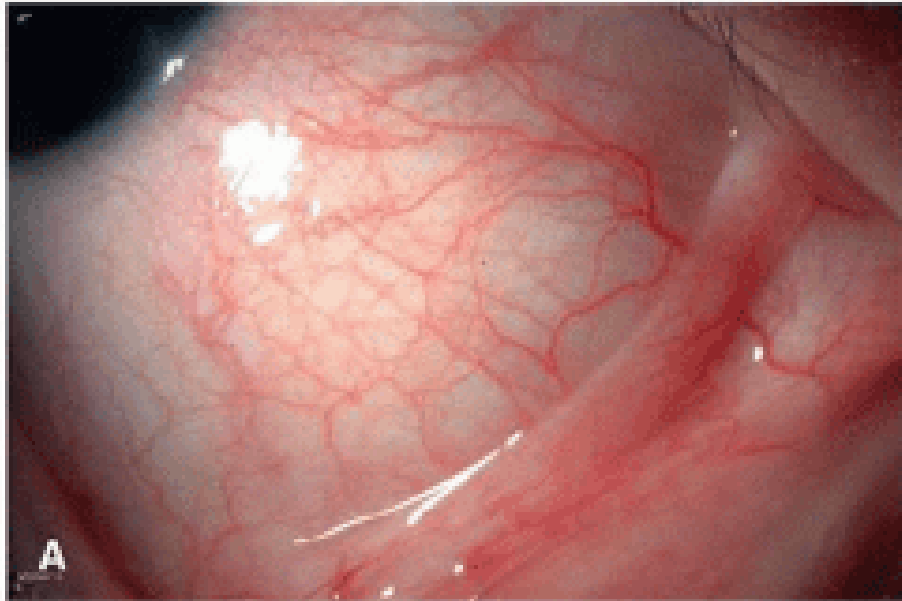
OS redness x 3/7

- Dull aching pain+
- No BOV

PMHx: unremarkable

Non-blanchable with 2.5%
phenylephrine





Phenylephrine test: blanching of blood vessels in episcleritis
(phenylephrine constricts superficial vessels only)



Scleritis

- Systemic associations in up to 50% (e.g. rheumatoid arthritis , relapsing polychondritis, Wegener's, Polyarthritidis nodosa)
- No known HLA association
- Other causes:
 - Infective (e.g. TB, syphilis)
 - Post surgery (e.g. pterygium surgery with beta-irradiation)



Systemic work-up

- Blood pressure, body weight (systemic immunosuppressants)
- Baseline CBC, L/RFT
- Inflammatory markers: ESR, rheumatoid factor (RA), ANA (lúpus), c-ANCA (Wegener's), p-ANCA (vasculitis, PAN)
- Hepatitis serology (steroid workup)
- VDRL (syphilis)
- Chest x-ray (steroid workup, TB)



Management

- Oral NSAIDs: first line
- Corticosteroids (1mg/kg/day)
- Immunosuppressants
- Biologics (e.g infliximab)



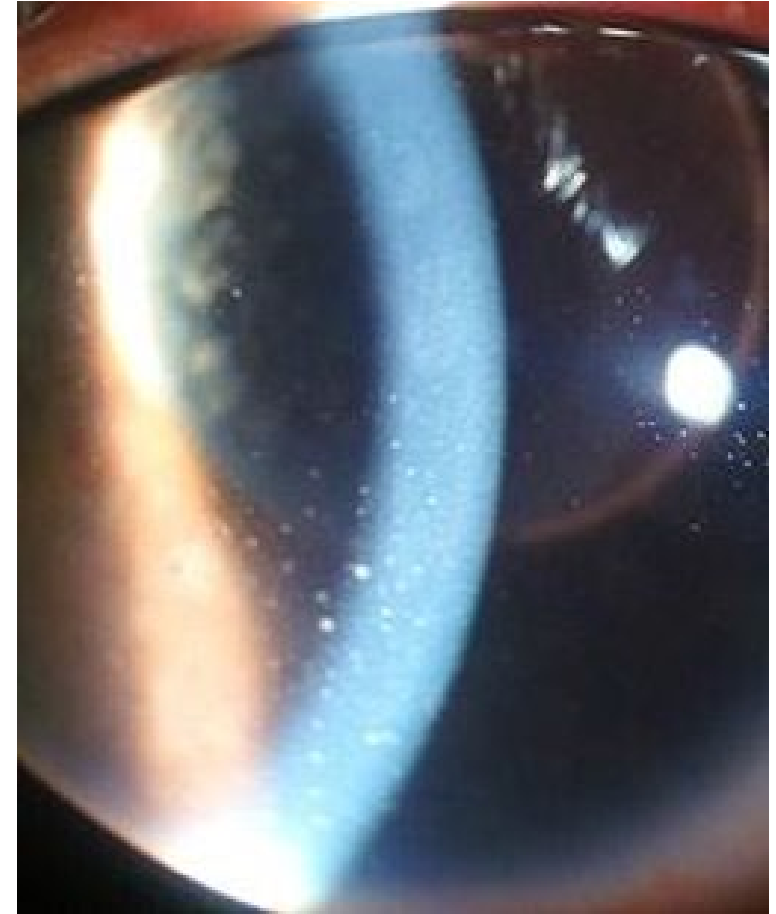
Case 6 25/M

OD redness x 3/7

- No pain
- Mild BOV+

PMHx

- Ankylosing spondylitis





Anterior uveitis

- **Classic signs:**
 - Ciliary flush
 - AC cells
 - Keratic precipitates
- **Treatment**
 - Steroids: topical, local injection, systemic
 - Cycloplegics



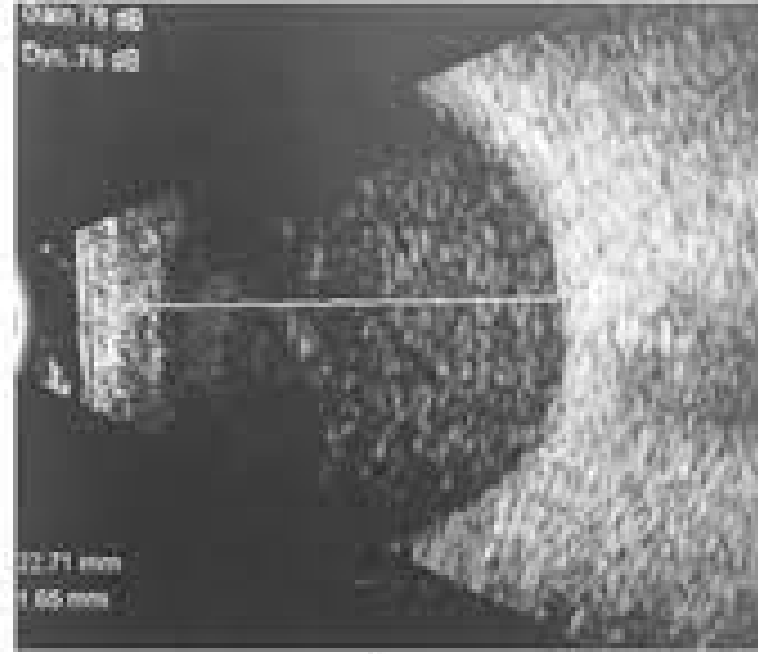
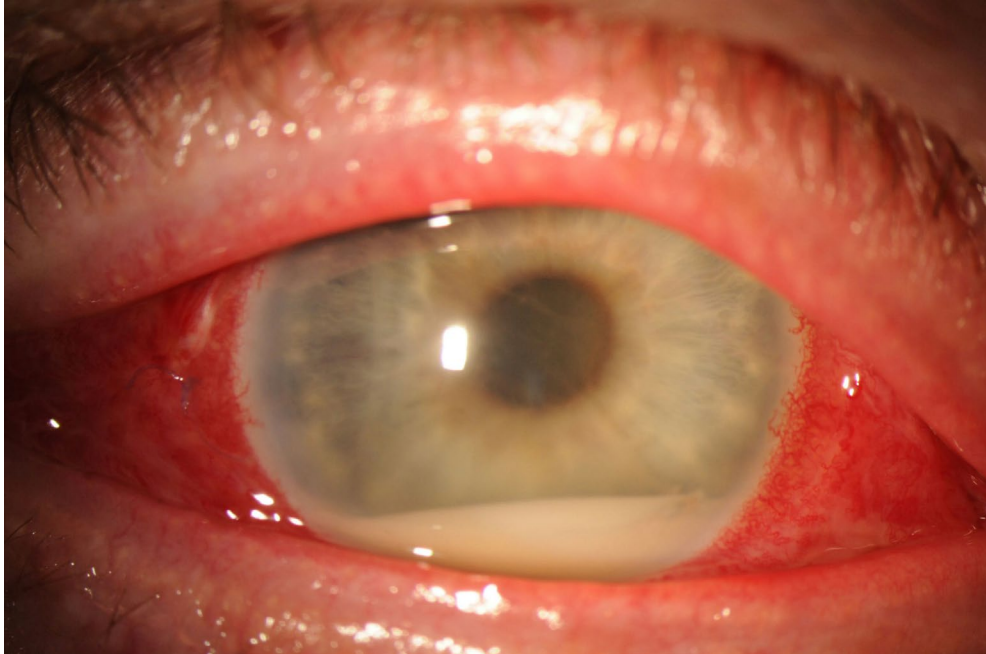
Systemic work-up for anterior uveitis

- No routine or standard work-up for anterior uveitis
- Many are isolated disease without systemic association
- Some commonly suggested tests
 - VDRL (syphilis)
 - TB (in endemic areas)
 - Lumbo-sacral spine x-ray, HLA-B27 (if suspect AS)



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Acute post-operative endophthalmitis



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RED FLAGS

- Ocular pain
- Decreased vision
- Photophobia
- Ciliary injection
- Corneal clouding/opacities
- Abnormal pupil
- High IOP

TELL US WHAT

SC Whole Class Session

YOU THINK

THANK YOU.



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)